



Introducción al análisis de datos NGS

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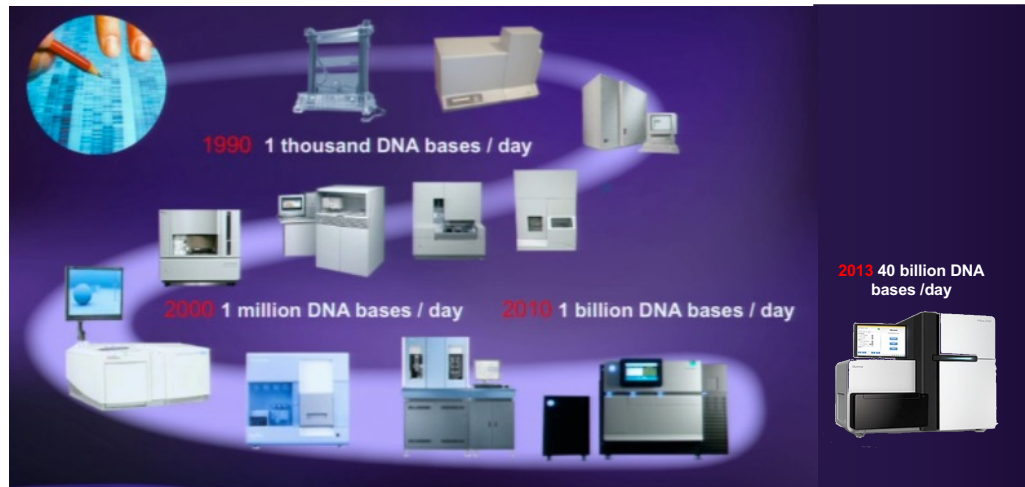
Temas a cubrir

1. Secuenciación por síntesis
2. Datos de secuenciación masiva
3. Flujos de trabajo NGS
4. Control de calidad de los datos

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Evolución de la técnica de secuenciación del ADN

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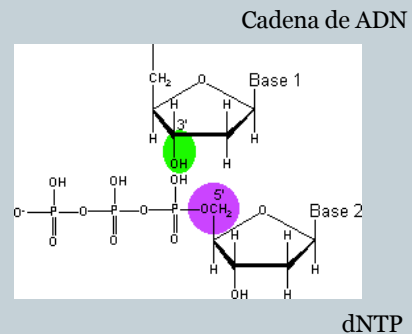
Introducción – Taller Genómica y Medicina – R. A. Verdugo - 2013

11/15/13

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Polimerización de los nucleótidos

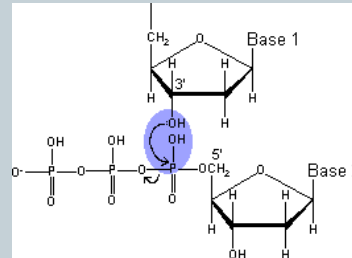
- El grupo 5' de un nucleótido trifosfato (dNTP) se acerca al grupo hidroxilo 3' de una cadena de nucleótidos.



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Polimerización de los nucleótidos

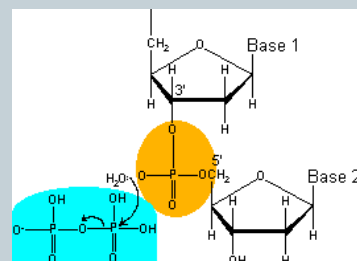
- El grupo hidroxilo 3' forma un enlace con el átomo de fósforo más próxima al átomo de oxígeno 5' del nucleótido libre (dNTP).
- El enlace entre el primer átomo de fósforo y el átomo de oxígeno que une a los próximos grupos fosfato se rompe.
- Se libera un protón (H^+) y un grupo OH^-



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Polimerización de los nucleótidos

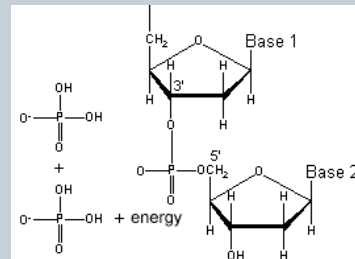
- Un nuevo enlace fosfodiéster une los dos nucleótidos
- Se libera un grupo pirofosfato
- Se libera un protón (H^+)



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Polimerización de los nucleótidos

- El grupo pirofosfato se hidroliza (se agregar agua)
- Se libera energía que se usa para la siguiente reacción



Más detalles en:

<http://www.ncbi.nlm.nih.gov/books/NBK22513/>

<https://www.chem.wisc.edu/deptfiles/genchem/netorial/modules/biomolecules/modules/dna1/dna13.htm>

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I. Secuenciación masiva

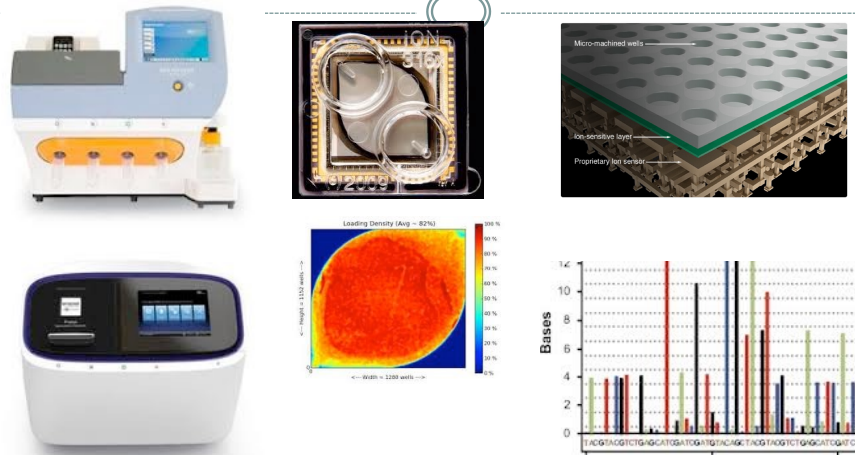
Secuenciación de última generación

Secuenciación paralela

Next Generation Sequencing (NGS)

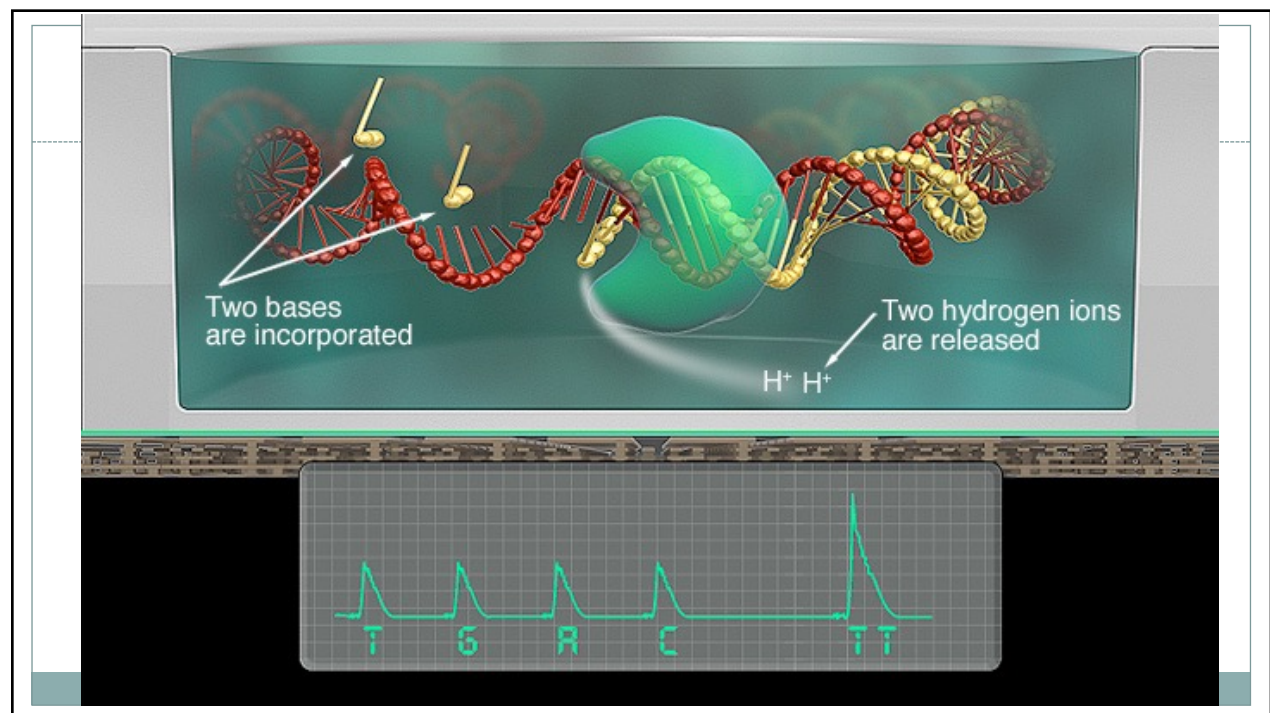
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Ion Torrent – Semi conductor seq



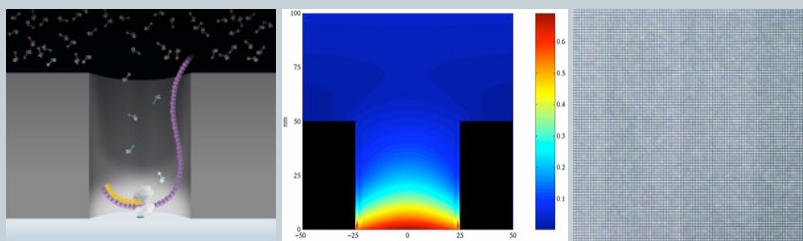
- Cada incorporación de un dNTP libera un ion H^+
- Detección por una placa detectora de iones
- Secuenciación por síntesis

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Pacific BioSciences

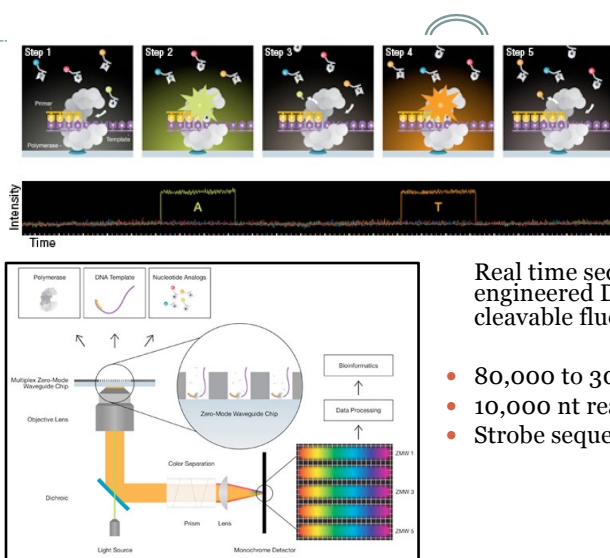


- Single Molecule, Real-Time (SMRT) Sequencing
- Average > 10,000 bp, some reads > 60,000 bp
- Single molecule sequencing suffers from poor signal/noise
- The Zero Mode Wave (ZMW) guide acts like a filter to restrict the fluorescence measurement to the well bottom



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Pacific BioSciences



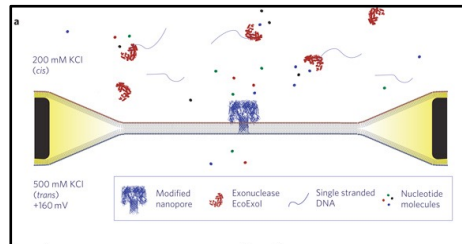
Real time sequencing thanks to engineered DNA polymerase and cleavable fluorophore.

- 80,000 to 300,000 wells.
- 10,000 nt read length
- Strobe sequencing possible

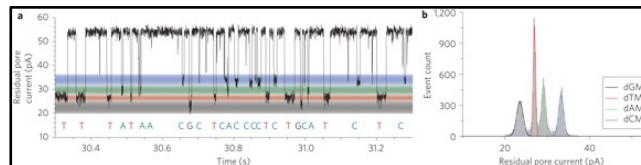


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Oxford Nanopore

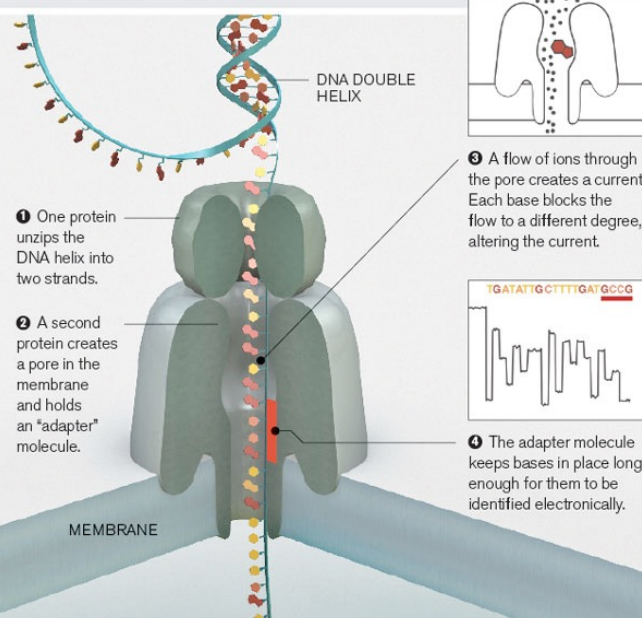


- Modified α Hemolysine with inserted Cyclodextrine
- Embedded in a lipid bilayer
- Flow of Nucleotide through the pore modifies the conductance
- Need to attach the Exonuclease to the pore



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DNA can be sequenced by threading it through a microscopic pore in a membrane. Bases are identified by the way they affect ions flowing through the pore from one side of the membrane to the other.

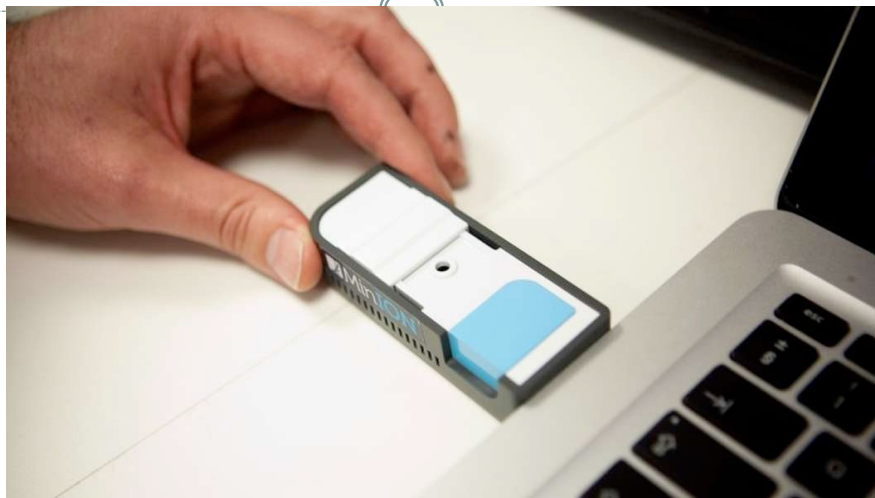


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MinION Oxford Nanopore Sequencer



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MinION : 512 pores (early access 2014) - disposable
GridION: 2000-8000 pores (TBD)
20 racks: human genome in 15 minutes



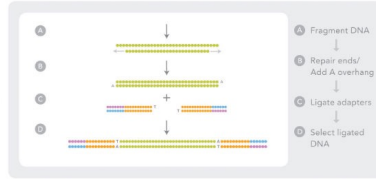
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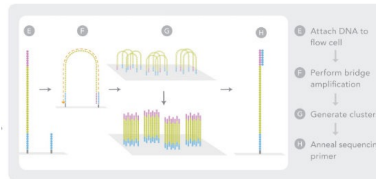
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Illumina Technology

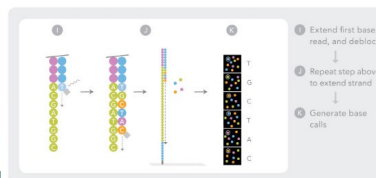
- Library Preparation



- Clonal amplification

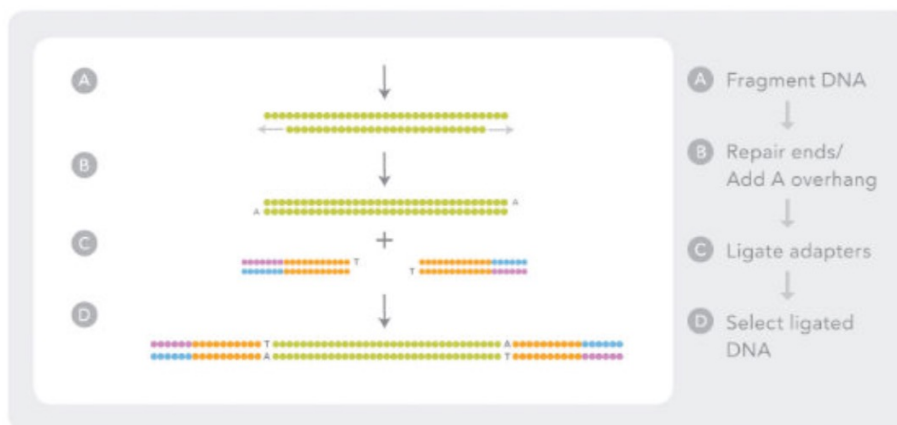


- Sequencing by synthesis



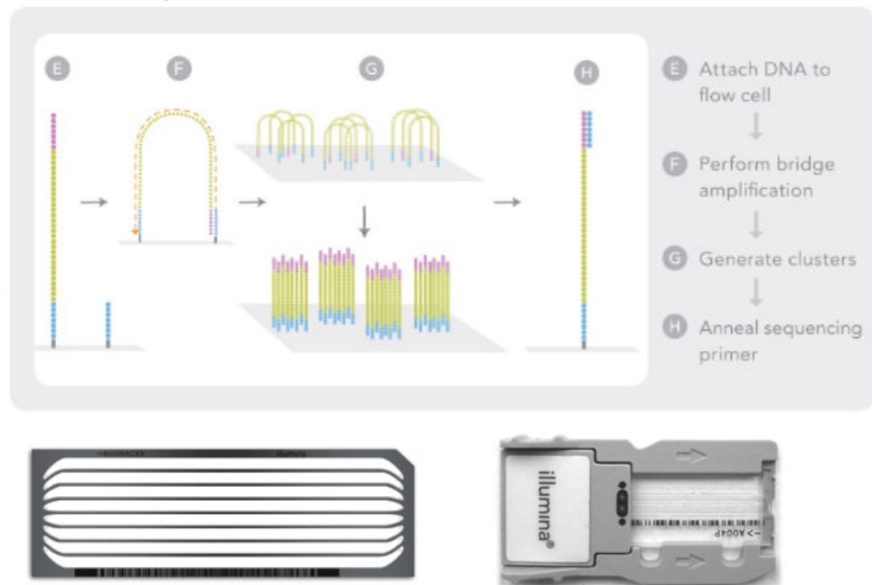
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- Library Preparation (TruSeq)



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Clonal amplification

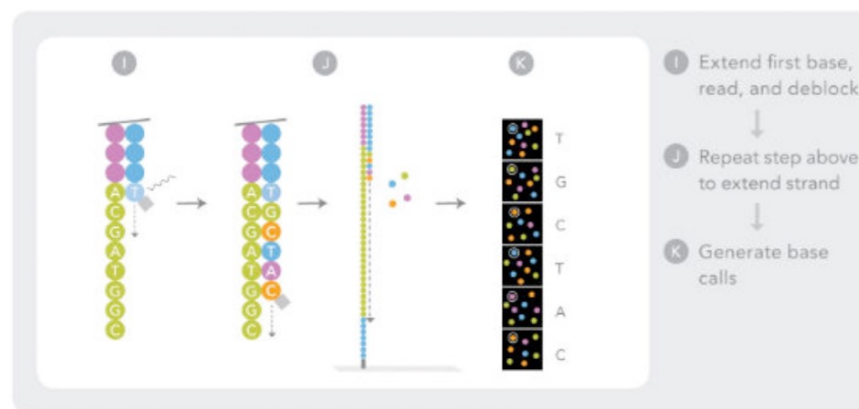


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Sequencing by synthesis



Secuenciación

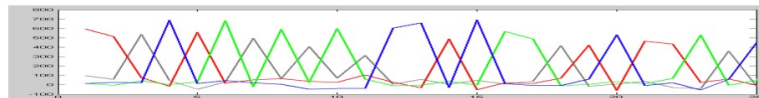
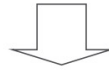
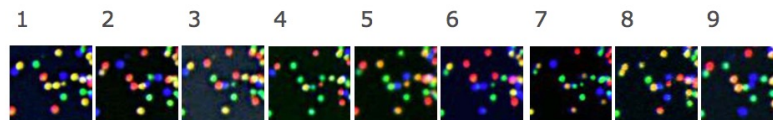
Más detalles en <http://bioinformaticsalgorithms.com/faqs/assembly.html>

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Image analysis identifies clusters (colonies, beads) and extracts intensity traces

- ▶ Detection: Find all clusters on the image
- ▶ Registration: Track clusters over multiple sequencing cycles
- ▶ Extraction: Provide intensity estimates for clusters in a given image



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Visualizing high-throughput Illumina four-color sequencing-by-synthesis

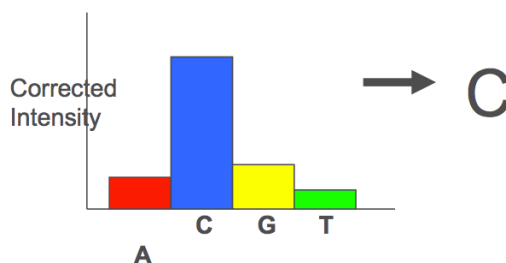
W. Trimble
Argonne National Laboratory

<http://bioinformatics.algorithms.com/faqs/assembly.html>

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Base-calling has two aspects: Identifying the base-call and assigning a confidence estimate to the call

- ▶ Making a base-call is usually based on the intensity estimates
 - Signal-processing needs to correct for confounding factors:
 - Frequency cross-talk (optical detection mechanism)
 - Phasing effects (imperfect chemistry)
 - Signal decay
- ▶ Assignment of a confidence estimate or quality score is vital for downstream analysis
 - phred method can be extended to Next_gen technologies



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Quality scores quantify the probability that a base-call is correct (or wrong)

Terminology:

- ▶ Base quality scores
 - Individual bases have quality scores which reflect the likelihood of the base being correct/incorrect
- ▶ Alignment scores
 - Probability that an alignment to a given position in the reference genome is correct
- ▶ Allele scores, SNP scores,
 - Probability that a given allele, SNP was observed (often conditional on the alignment being correct)
- ▶ Base and alignment scores are single read scores; SNP scores are consensus scores
 - Consensus calls use information from multiple reads

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Phred scores

1. A base quality score assigned by the phred software (or a program based on the phred)
2. A quality score expressed on a logarithmic scale:
 - ▶ $Q = -10 \log_{10}(\text{probability of an error})$
 - ▶ Example: Q20 = 1% error probability

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Capacidad de secuenciación de equipos Illumina disponibles en el mercado



MiSeq



NextSeq 500



HiSeq 2500

	N/A	Mid Output	High Output	Rapid Run	High Output
Flow cells processed per run	1	1	1	1 or 2	1 or 2
Output range	0.5 - 15 Gb	20 - 39 Gb	30 - 120 Gb	10 - 180 Gb	50 - 1000 Gb
Run time	5 - 65 hours	15 - 26 hours	12 - 30 hours	7 - 40 hours	< 1 day - 6 days
Single-end reads per flow cell	25 Million	130 Million	400 Million	300 Million	2 Billion
Max read length	2 × 300 bp	2 × 150 bp	2 × 150 bp	2 × 150 bp	2 × 125 bp

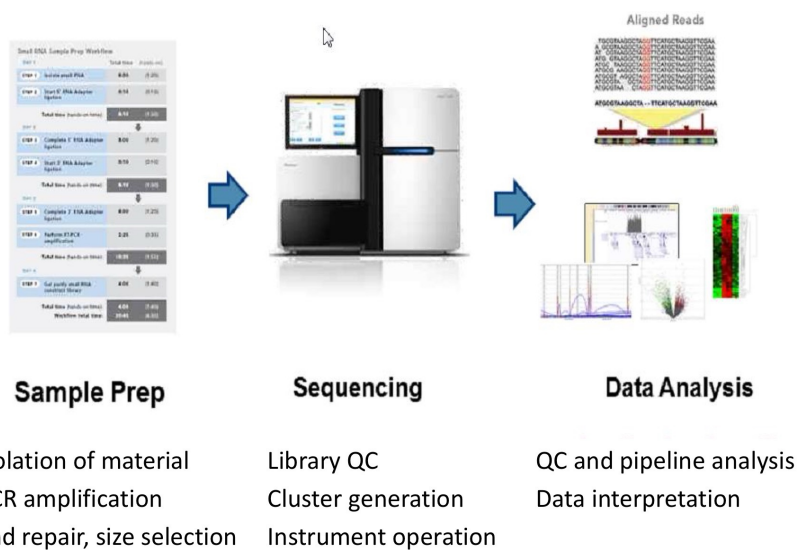
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Número de muestras que pueden ser estudiadas en una corrida

Key Area	MiSeq	NextSeq 500	HiSeq 2500
DNA Sequencing			
Whole Genome - Large (e.g. Human)	-	1	1-10
Whole Genome - Small (e.g. E. coli)	1 - 96	120 - 792	96 - 6660
Targeted - Exome or large panels	1	1 - 12	12 - 160
Targeted - Small gene panels	3	15 - 48	36 - 72
RNA Sequencing			
RNA Profiling	1 - 2	12 - 36	24 - 396
Transcriptome Analysis	-	3 - 10	8 - 96
Small RNA Analysis	1 - 5	25 - 80	60 - 792
Targeted RNA	384	384	6144
Regulation Applications			
ChIP-Seq	1	8 - 24	20 - 264
Methylation Analysis	-	1	

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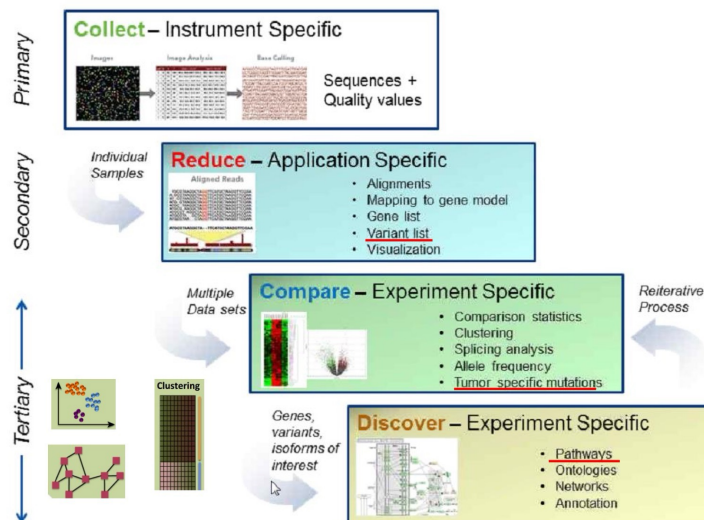
II. Basic NGS Workflow



Olson et al.

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High Throughput Data Analysis Overview

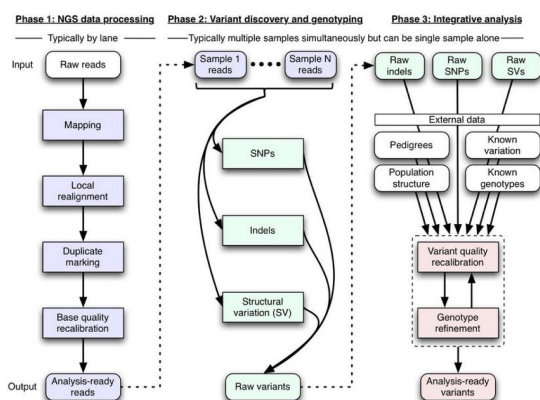


Olson et al.

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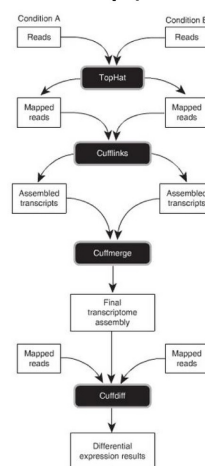
Typical Data Analysis Pipelines

Genotyping (GATK)



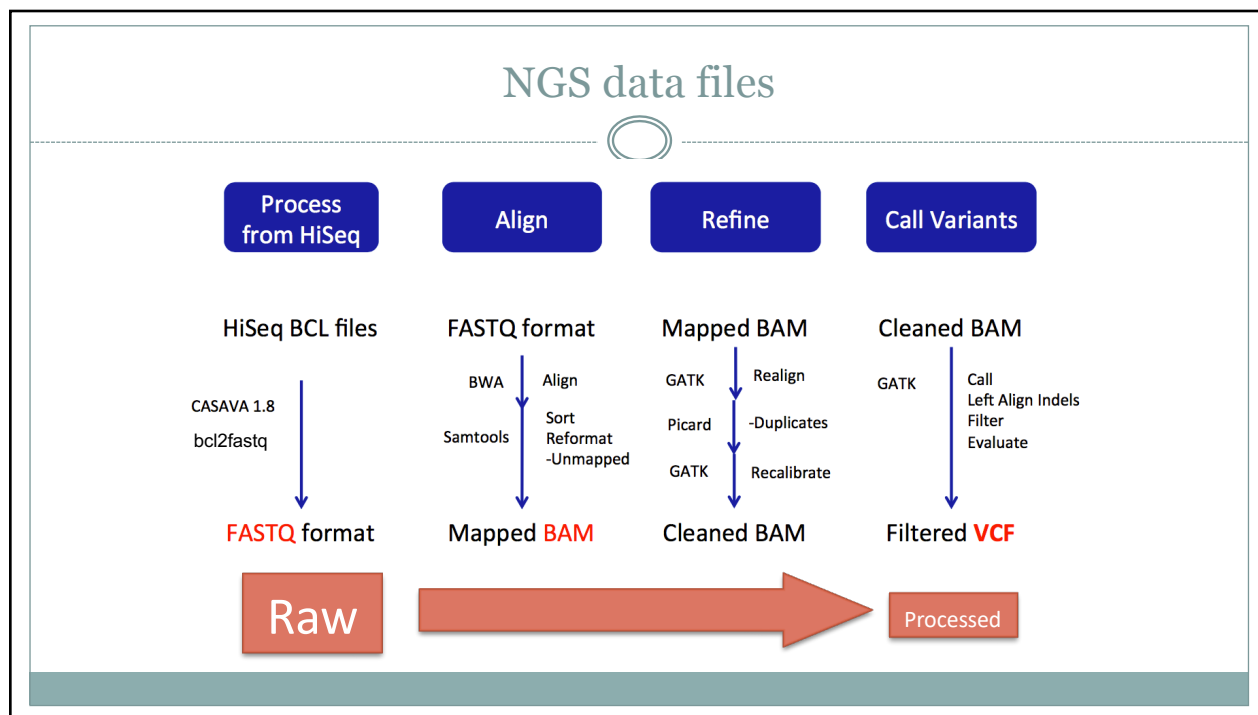
http://www.broadinstitute.org/gsa/wiki/images/7/7a/Overall_flow.jpg
<http://www.broadinstitute.org/gatk/guide/topic?name=intro>

RNA-seq (Tuxedo)



<http://www.nature.com/nprot/journal/v7/n3/full/nprot.2012.016.html>

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File Sizes

	Depth	FASTQ.gz	BAM	VCF.gz
Genome	30x	100-200Gb	100-200 Gb	2Gb
Exome (50 Mb)	120x	5-10 Gb	10-30Gb	3-6Mb
Deep Targeted (150 kb)	1500x	300-500Mb	150Mb	3-6Mb

Assume 100 bp read length

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Raw sequencing data: Fastq format

```
@IL31_4368:1:1:996:8507:2
TCCCTTACCCCAAGCTCCATACCTCCTAATGCCACACCTCTTACCTTAGGA
+
FFCEFFFEFFFEFFFEFFFEFFFCFC<EEFEFFFCFF<;EEFF=FEE?FCE
@IL31_4368:1:1:996:21421/2
CAAAAACCTTCACCTTACCTGCCGGGTTCCAGTTTACATTCCTACTGTTTGAC
+
>DBDDb,B9BAA4AAB7BB?7BBB=91;+*@;5<87+*=/*@@?9=73=.7)7*
@IL31_4368:1:1:997:10572/2
GATCTTCTGTGACTGGAAGAAAATGTGTACATATTACATTCTGTCCCCATTG
+
E?~EECE<EEEE98EEEEAEED?BE@AEAB><EEABCEDEC<<EBDA=DEE
@IL31_4368:1:1:997:15684/2
CAGCCTCAGATTACGATTCCTCAAATTCAGCTGCCGGCTGAAACAGCAGCAGAC
+
EEEEEEE9EAEEDEEEEEEEEECEAAEED<CD=D*BCAC?;CB,<D@,
@IL31_4368:1:1:997:15249/2
AATGTTCTGAAACCTCTGAGAAGCAAATATTTATTTAATGAAAAATCCTTAT
+
EDEEC;EEE;EEE?EECE;7AEEEEEE0EECEA;D6D>+EE4E7EEE4;E=EA
@IL31_4368:1:1:997:6273/2
ACATTACCAAGACCAAGGAACTTACCTTGCAAGAATTAGACAGTTTCATTG
+
EAAAFFFEFFFCFAFFAFCCFFFEFF>EFFFFB?ABA@ECEE=<F@DE@DDF;
@IL31_4368:1:1:997:1657/2
CCCACTCTCTCAATGTTTCCATATGGCAGGACTCAGCACAGTGGAATTAAT
(...)
```

- Instrument serial #
- Lane
- Swath
- X coord
- Y coord
- Read direction

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Fastq Read ID Variations

@EAS139:136:FC706VJ:2:2104:15343:197393 1:Y:18:ATCACG

EAS139	the unique instrument name
136	the run id
FC706VJ	the flowcell id
2	flowcell lane
2104	tile number within the flowcell lane
15343	'x'-coordinate of the cluster within the tile
197393	'y'-coordinate of the cluster within the tile
1	the member of a pair, 1 or 2 (<i>paired-end or mate-pair reads only</i>)
Y	Y if the read is filtered, N otherwise
18	0 when none of the control bits are on, otherwise it is an even number
ATCACG	index sequence

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IV. Control de Calidad en NGS

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1. Calidad de la muestra de ADN o ARN

- Revisar degradación
- Espectrofotometría (Nanodrop)
- Fluorimetría (Pico- y Ribo-Green)
- Electroforesis en gel o capilar (Bioanalyser, TapeStation, Fragment Analyzer). RNA Integrity Number (RIN) de Agilent para RNA (RIN>7)

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Control de Calidad en NGS

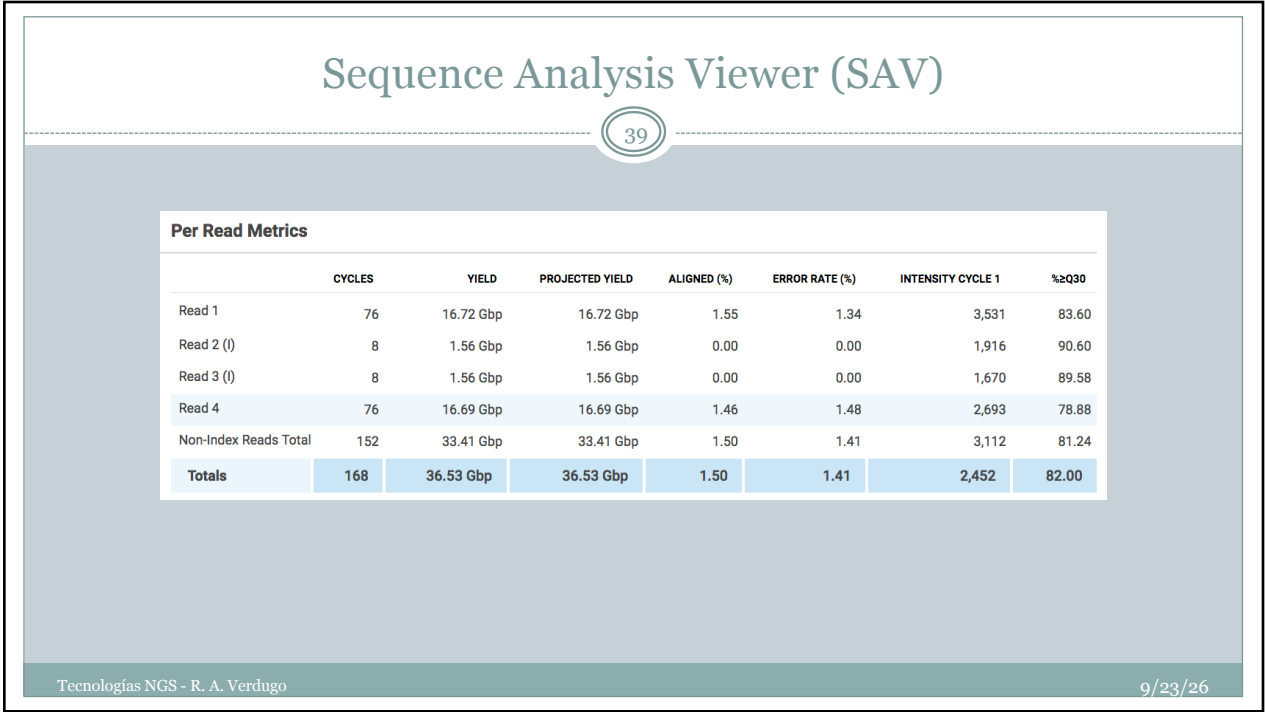
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1. Calidad de las secuencias

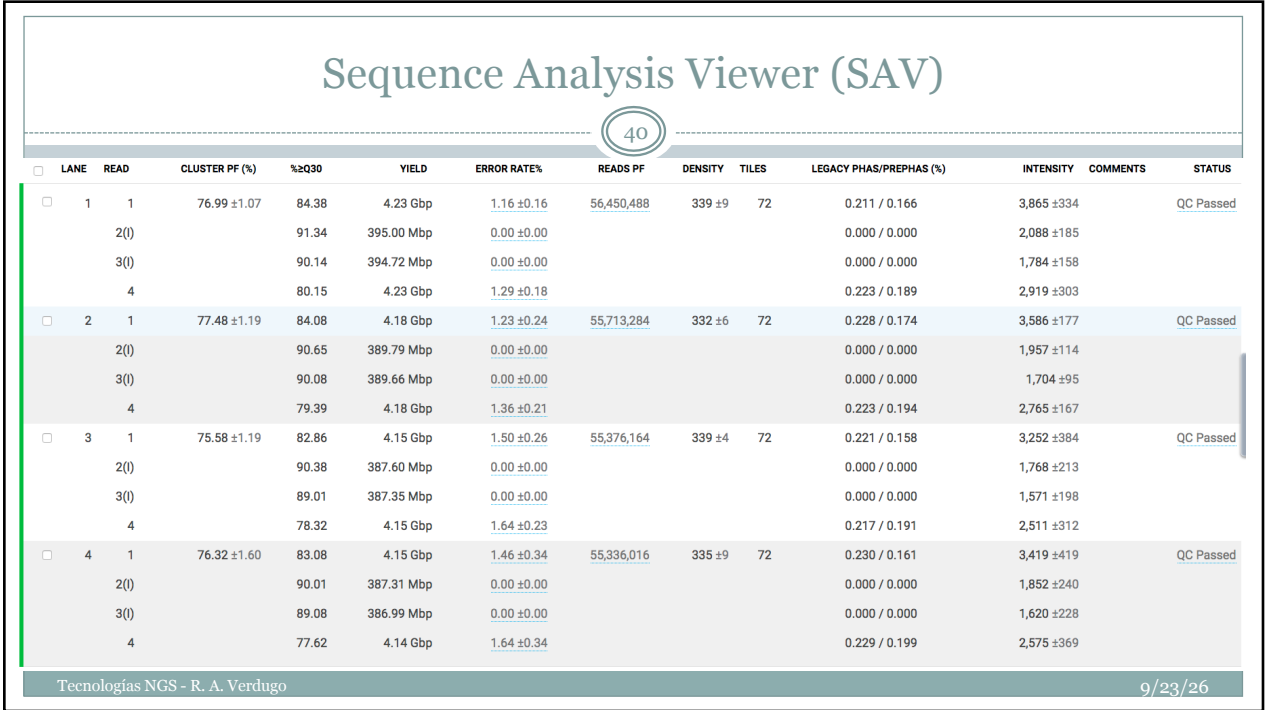
- Sequence Analysis Viewer (SAV) en Basespace
- FastQC¹
- MultiQC
- FQC Dashboard (*Bioinformatics*:33(19):3137)

¹<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/Help/3%20Analysis%20Modules/>

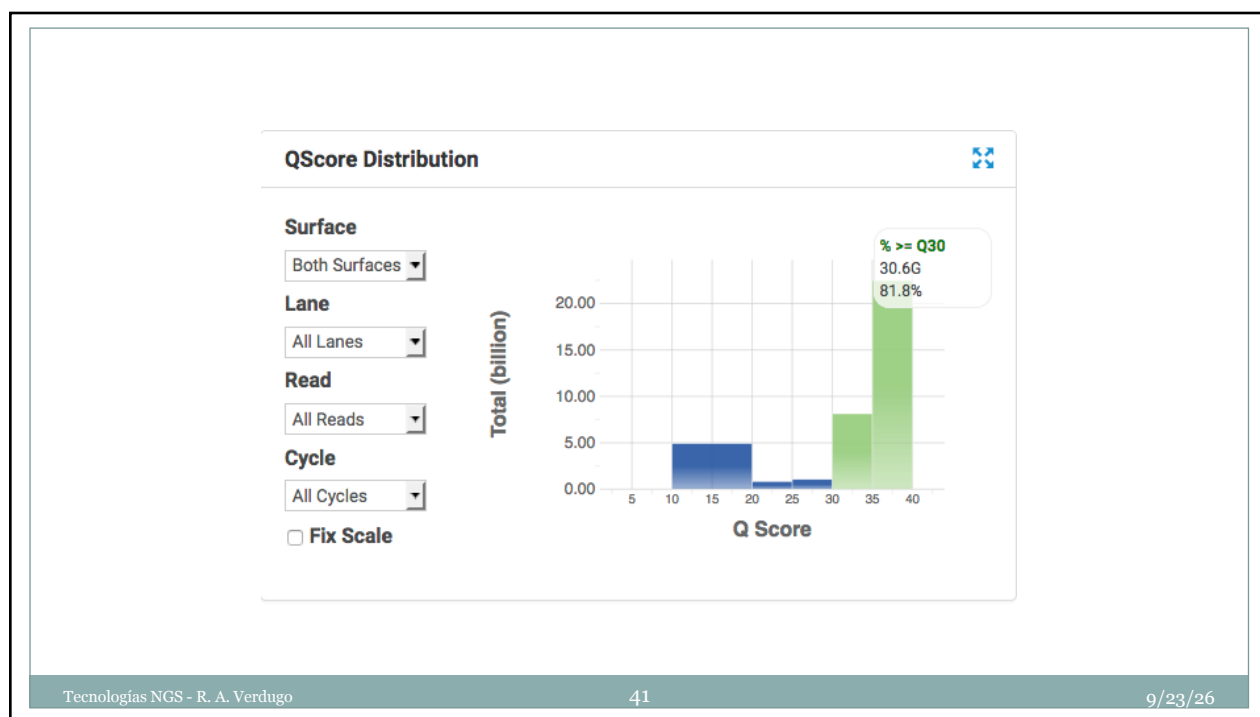
38



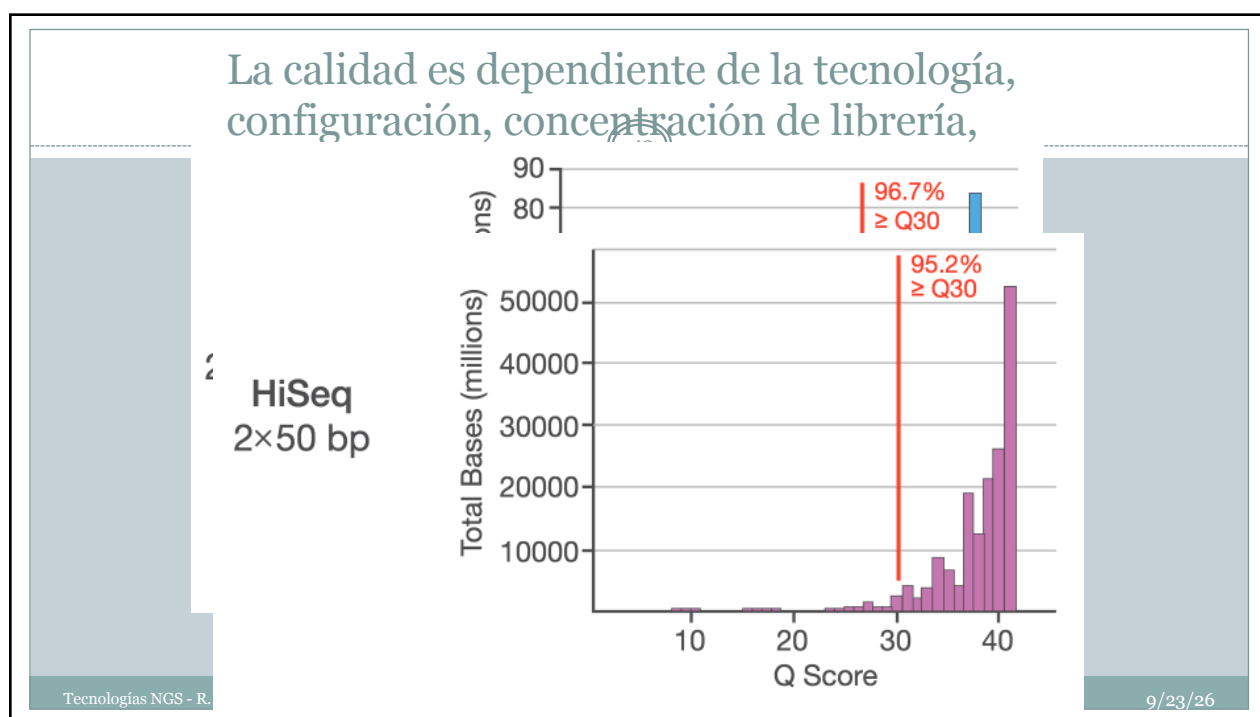
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Calidad Q30 en MiSeq

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Quality Scores[†]

MiSeq Reagent Kit v2	MiSeq Reagent Kit v3
> 90% bases higher than Q30 at 1 × 36 bp	> 85% bases higher than Q30 at 2 × 75 bp
> 90% bases higher than Q30 at 2 × 25 bp	> 70% bases higher than Q30 at 2 × 300 bp
> 80% bases higher than Q30 at 2 × 150 bp	
> 75% bases higher than Q30 at 2 × 250 bp	

† A quality score (Q-score) is a prediction of the probability of an error in base calling. The percentage of bases > Q30 is averaged across the entire run.

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Calidad Q30 en NextSeq

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Quality Scores^{††}

NextSeq 550 System High-Output Kit	NextSeq 550 System Mid-Output Kit
> 75% bases higher than Q30 at 2 × 150 bp	> 75% bases higher than Q30 at 2 × 150 bp
> 80% bases higher than Q30 at 2 × 75 bp	> 80% bases higher than Q30 at 2 × 75 bp
> 80% bases higher than Q30 at 1 × 75 bp	

†† A quality score (Q-score) is a prediction of the probability of an error in base calling. The percentage of bases > Q30 is averaged across the entire run.

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Valores históricos de calidad obtenidos en ChileGenómico

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☐	RUN NAME	INSTRUMENT FLOWCELL ID	% PF	AVG %Q30	YIELD	OWNER	CREATED	SIZE	CYCLES	LANE & QC STATUS	RUN STATUS
☐	WESSGPooling3	NB551... HGLK5A...	76.60%	82.00%	36.53 Gbp	Ricardo ...	2017-11-2...	22 GB	76 76	QC Passed	Complete
☐	Pool4MomiNiñoelPlomo	NB551... H3CY2B...	97.07%	96.91%	12.54 Gbp	Ricardo ...	2017-11-0...	7 GB	38 38	QC Passed	Failed
☐	SEQMomi_DNAAntiguo	M0315... 0000000...	97.43%	90.52%	1.36 Gbp	Ricardo ...	2017-09-2...	2 GB	76 76	QC Passed	Complete
☐	PoolExomeAgilentRun3	NB551... HGL3NA...	87.40%	88.27%	26.64 Gbp	Ricardo ...	2017-08-0...	14 GB	76 76	QC Passed	Complete
☐	WGSINIAII	NB551... H5VC2B...	93.49%	83.12%	130.91 Gbp	Ricardo ...	2017-06-1...	64 GB	151 151	QC Passed	Complete
☐	Poolprunus2	NB551... HG2LYB...	94.35%	85.25%	122.51 Gbp	Ricardo ...	2017-06-0...	59 GB	151 151	QC Passed	Complete
☐	AgilentExomasrun2	NB551... HGLM3A...	83.80%	87.95%	35.23 Gbp	Ricardo ...	2017-05-2...	19 GB	76 76	QC Passed	Complete
☐	AGILENTEXOME1	NB551... HGL3YA...	86.13%	88.59%	29.42 Gbp	Ricardo ...	2017-05-2...	16 GB	76 76	QC Passed	Complete
☐	EDP4-Run2	NB551... HGL5HA...	77.39%	84.30%	32.48 Gbp	Ricardo ...	2017-05-1...	1 GB	76 76	QC Passed	Complete
☐	Patagonia_WG_Run_1	NB551... HWHYGB...	95.43%	88.16%	127.13 Gbp	Ricardo ...	2016-04-2...	59 GB	151 151	QC Passed	Complete

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Data By Cycle

Chart

Intensity

Surface

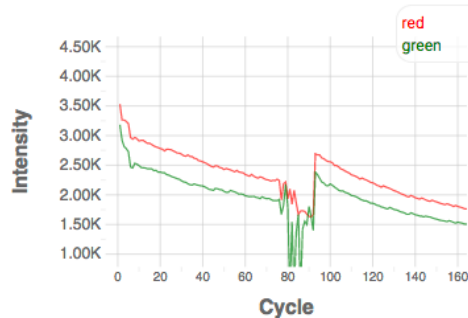
Both Surfaces

Lane

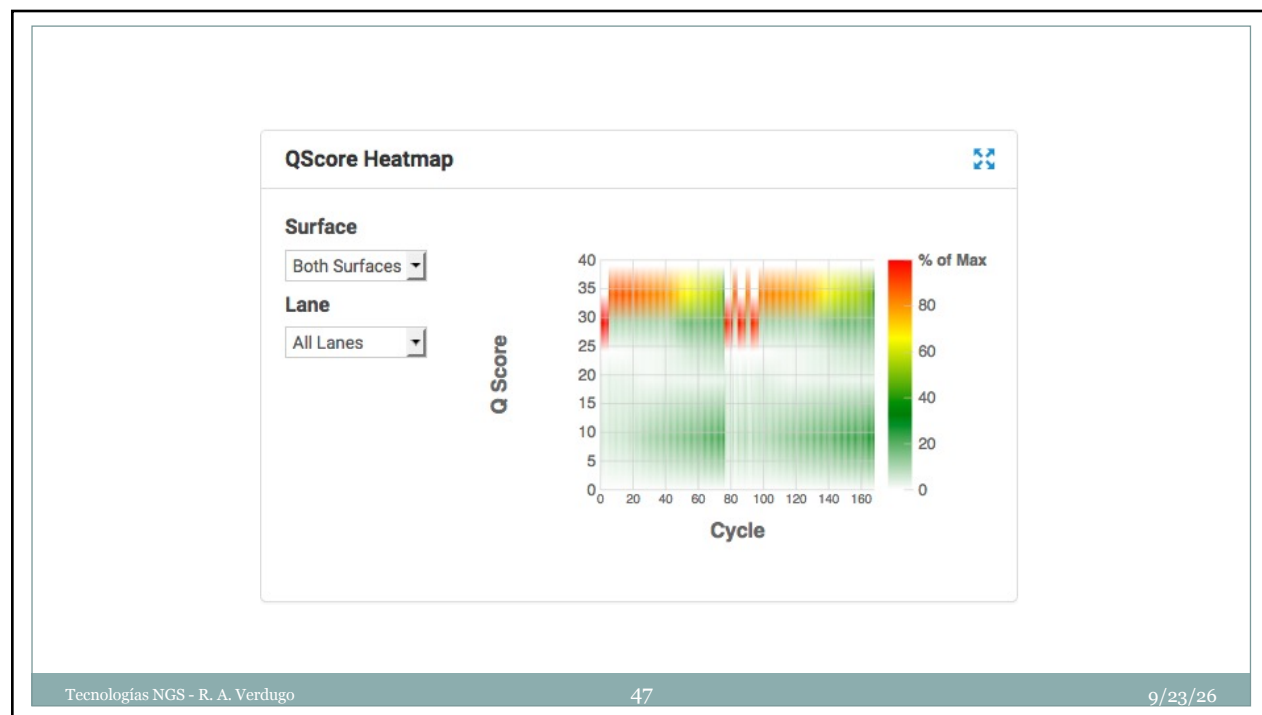
All Lanes

Channel

All Channels

☐ Fix Scale

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Densidad de Clusters

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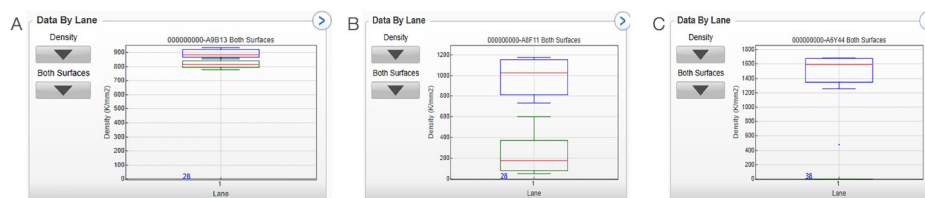
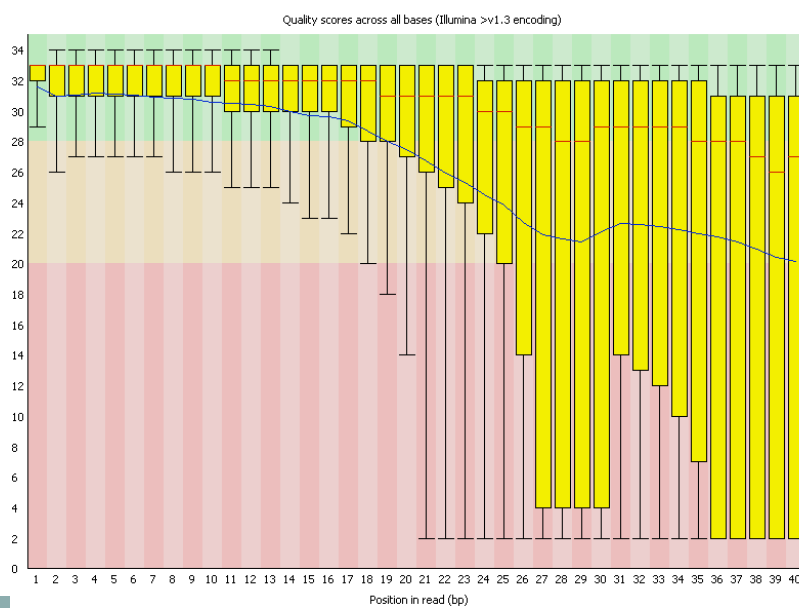


Figure 3: Data by Lane: Density. The blue boxes illustrate the raw cluster density range, the green boxes illustrate the %PF cluster density range, and the red lines indicate the median cluster density values. A) Optimal density. B) Overclustered. C) Severely overclustered.

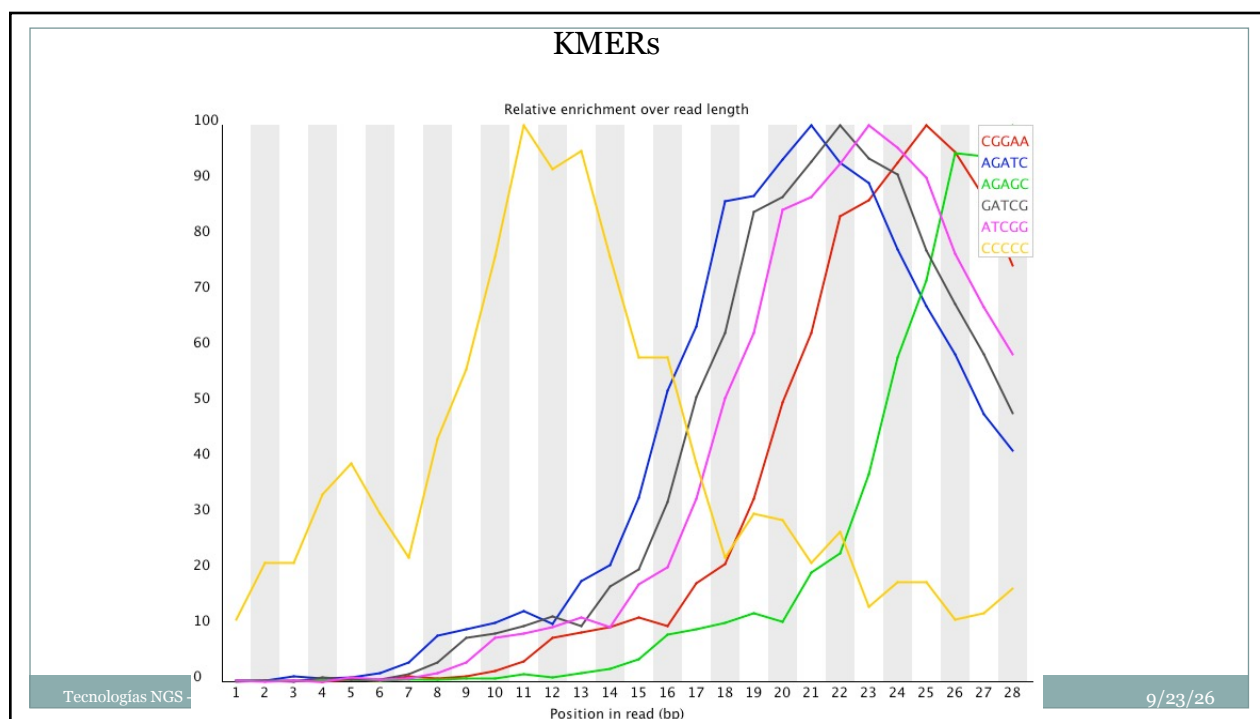
Table 1: Optimal Raw Densities for Illumina sequencing systems

	MiniSeq™	MiSeq		NextSeq®	HiSeq 2500 Rapid Run (RR)	HiSeq 2500 High Output (HO)	
Versions	High and Mid Output	v2	v3	v2 High and Mid Output	v1 and v2	v3	v4
Raw Density (K/mm²)	170–220	1000–1200	1200–1400	170–220	850–1000	750–850	950–1050

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